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Multi-level tray loader

Abstract

A system and method are provided for loading trays with articles using a first tray conveyor on which empty trays are delivered and an article conveyor on which articles are delivered to be transferred onto empty trays. The trays include a frame with a peripheral raised edge and a base having at least one aperture and a transport plate for the articles, the transport trays being positioned on the base of the frame. A lifting apparatus is provided for the transport plate of the tray at a height that is about equal to the height of the article conveyor, such as about 10 millimeters (10 mm) or less to reduce or eliminate drop forces as articles as transferred from the article conveyor onto the tray.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Prov. Pat. Appl. Ser. No. 63/334,834, filed Apr. 26, 2022.

FIELD OF THE INVENTION
(1) The present invention is directed to a material handling system, and in particular a fragile article handling system.
BACKGROUND OF THE INVENTION
(2) The invention relates to systems and methods for loading trays onto articles in a conveyor system. Systems and methods of this type are used for automatic loading of trays loaded with articles, wherein the loading process is carried out during ongoing conveyor movement, incorporated into supply and discharge conveyor technology. Trays are often used in automatic storage and picking systems. These trays allow articles, which are often highly dissimilar in their dimensions and/or material properties, to be handled in logistics systems quickly and in a controlled and uniform manner in spite of this dissimilarity in the articles.
SUMMARY OF THE INVENTION
(3) The present invention provides a tray loading system for loading trays with fragile product (i.e. fresh produce, such as tomatoes, glass products, and other products poorly suited for drop impacts) onto trays and thus enables the loading of hard to handle and/or fragile products in a material handling system.
(4) According to one form of the present invention, a tray loading system is provided for loading articles onto trays in a material handling conveyor system. The tray loading system includes a first or upstream tray conveyor for transporting material handling transport trays. The trays utilized in the material handling conveyor system each include a frame having a peripheral raised edge and a base having at least one aperture. A transport plate with a transport surface for supporting articles is supportable on the base of the tray frame. The transport plate and is moveable relative to the raised edge and base of the frame. The system includes a transfer bed downstream of the first tray conveyor and configured to receive a tray from the first tray conveyor. The system also includes an article conveyor upstream of the transfer bed and adjacent to the first tray conveyor, such as above or alongside the first tray conveyor. The article conveyor transports articles nearby the transfer bed where the article is then transferrable from the conveyor surface of the article conveyor onto the transport surface of a transport tray. The system includes a second tray conveyor downstream of the transfer bed and operable to transport trays away from the transfer bed. The transfer bed is operable to position the tray in a substantially level orientation in which the transport plate of the tray is at or above the raised edge of the tray frame such that the transport surface of the transport plate is substantially parallel with the conveyor surface of the article conveyor. The transfer bed is operable to position the tray such that the base of the tray is substantially parallel to a conveyor surface of a downstream tray conveyor.
(5) In one aspect, the transfer bed is pivotable about a horizontal axis and operable to move between: (i) a horizontal orientation in which the transport plate of a tray supported at the transfer bed is horizontal and substantially parallel with the conveyor surface of the article conveyor to receive an article from the article conveyor with about five millimeters (5 mm) or less drop to the

transport plate; and (ii) an oblique orientation in which the transport plate of a tray supported at the transfer bed is obliquely oriented relative to horizontal and substantially parallel with the conveyor surface of the second tray conveyor to transfer the tray from the transfer bed to the second tray conveyor.
(6) In another aspect, the transfer bed includes a lift apparatus to position the transport plate of a horizontally orientated tray supported on a transport surface of the transfer bed at or above the peripheral raised edge of the frame of the tray. The lift apparatus may include a plurality of support elements fixed relative to the first tray conveyor and the article conveyor and configured to reach through the transport surface of the transfer bed and through the aperture in the tray as the frame of the tray is lowered relative to the support elements. As such, the frame lowers relative to the transport plate of the tray.
(7) In yet another aspect, the transfer bed includes a lift apparatus operable to lift the transport plate of a horizontally orientated tray supported on the transfer bed to a position substantially coplanar to the conveyor surface of the article conveyor such that the transport plate is positioned to receive an article from the article conveyor with five millimeters (5 mm) or less drop to the transport plate. The lift apparatus includes a plurality of vertically moveable support elements operable to reach through the transport surface of the transfer bed and the aperture in the tray such that as the plurality of support elements raise, they lift the transport plate of the tray supported at the transfer bed above the peripheral raised edge of the tray and the frame of the tray remains on the transport surface of the transfer bed.
(8) In another form of the present invention, a method is provided for loading a material handling transport tray in a material handling conveyor system. The method includes providing a first tray conveyor, an article conveyor adjacent the first tray conveyor, a tray, and a transfer bed to receive the tray from the first tray conveyor. The tray includes a frame having a peripheral raised edge and a base having at least one aperture, and a transport plate with a transport surface for supporting articles, where the transport plate is supportable on the base of the frame for transport and is moveable relative to the raised edge and base of the frame. The method includes delivering a tray with the first tray conveyor to the transfer bed and then positioning the transport plate of the tray at the transfer bed such that the transport surface of the transport plate is less than about ten millimeters (10 mm) below the conveyor surface of the article conveyor. An article is transferred from the conveyor surface of the article conveyor to the transport surface of the transport plate of the tray at the transfer bed. The loaded tray is transferred away from the transfer bed. The loaded tray may be transferred to a second tray conveyor downstream of the transfer bed.
(9) Accordingly, systems and methods are provided for loading fragile articles onto trays with minimal drop distances such that the articles experience minimal impact, which may otherwise damage the articles, such as fragile produce, for example. The systems include lifting apparatuses to position movable transport plates of trays at a height that is at or about equal to the height of a supply conveyor delivering the articles to the tray.
(10) These and other objects, advantages, purposes, and features of this invention will become apparent upon review of the following specification in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a side perspective view of a plurality of tray loading systems for a material handling facility, in accordance with the present invention, with two tray loading systems in an oblique orientation and two tray loading systems in a horizontal orientation;
(2) FIG. **2** is a side elevation view of the tray loading systems of FIG. **1**;
(3) FIG. **3** is another side elevation view of the tray loading systems of FIG. **1**, depicted with all of

the tray loading systems in a horizontal orientation;
(4) FIG. **4** is a top plan view of the tray loading systems of FIG. **1**;
(5) FIG. **5** is a top side perspective view of a transport tray for use in a material handling system;
(6) FIG. **6** is another top side perspective view of the transport tray of FIG. **5**;
(7) FIG. **7** is a bottom perspective view of the transport tray of FIG. **5**;
(8) FIG. **8** is a side elevation view of another tray loading system in accordance with the present invention, depicted with a lift apparatus in a lifted or raised position;
(9) FIG. **9** is a top plan view of a plurality of the tray loading systems of FIG. **8**;
(10) FIG. **10** is a side elevation view of another tray loading system in accordance with the present invention, depicted with a lift apparatus in a lifted or raised position;
(11) FIG. **11** is a top plan view of a plurality of the tray loading systems of FIG. **10**;
(12) FIG. **12** is a front side perspective view of another tray loading system in accordance with the present invention, depicted with a pair of lateral pushers in a retracted position;
(13) FIG. **13** is a rear side perspective view of the tray loading system of FIG. **12**, depicted with a downstream one of the pair of lateral pushers in an extended position;
(14) FIG. **13**A is a plan view of the loading system of FIGS. **12** and **13** showing the upstream and downstream conveyors; and
(15) FIG. **14** is a diagram of a method for loading articles onto a tray, in accordance with the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(16) Referring now to the drawings and the illustrative embodiments depicted therein, a tray loading system is provided for material handling systems that utilize transport trays with false bottoms. The tray loading system is provided for reducing or eliminating impacts to fragile articles or items, such as soft produce or glass products, as the items are loaded onto a transport tray. The configuration of the tray loading system enables the transfer of items from an article conveyor to a transport tray with a minimal or very small amount of drop from the article conveyor to a transport surface of the transport tray. Various configurations of the tray loading system are contemplated, including configurations having pivotable tray transfer beds, a tray transfer bed with a vertical lifting apparatus or lift, and a lateral transfer apparatus or pusher to move articles toward a tray transfer bed.
(17) Referring to the illustrative embodiments of FIGS. **1-4**, a tray loading system **100** is provided for a material handling system **102** having a first or upstream tray conveyor **104** for transporting material handling transport trays **106**, a second or downstream tray conveyor **108** for transporting trays **106**, and an upstream article conveyor **110** for transporting articles or items **112**, such as fragile produce or glass products. The tray loading system **100** includes a transfer bed **114** for facilitating a transfer of an article **112** from the article conveyor **110** to a tray **106** supported at the transport surface of the transfer bed **114**. The upstream tray conveyor **104** delivers trays **106** to the transfer bed **114** and the downstream tray conveyor **108** transports trays **106** away from the transfer bed **114**.
(18) The trays **106** utilized in the material handling system **102** may have structure that is similar or substantially identical to and function in similar manner as the trays disclosed in commonly owned and assigned U.S. Pat. No. 8,763,785, issued Jul. 1, 2014 to Dematic GmbH of Offenbach, Germany, and/or commonly owned and assigned U.S. Patent Application Pub. No. 2022/0055843A1, which was published on Feb. 24, 2022, the disclosures of which is hereby incorporated herein by reference in their entireties. Referring to FIGS. **5-7**, the trays **106** each include a frame **116** provided with a peripheral raised edge **118** and a base **120** having a central aperture **122** extending in the longitudinal direction. The tray **106** has a comparatively thin flat transport plate **124** for supporting articles **112**. The plate **124** lies on or is supportable at the base **120**, such that the plate **124** is moveable relative to the frame **116**. The plate **124** includes a transport surface **124***a* for supporting articles **112**.

(19) The transfer bed **114** is positioned downstream of the upstream tray conveyer **104** and includes a transport or support surface **104***a* for supporting the frame **116** of a tray **106** at the transfer bed **114**. In the illustrated embodiment of FIGS. **1-4**, the upstream tray conveyor **104** is oriented at an oblique angle relative to horizontal, such that a tray **106** on conveyer **104** approaches the transfer bed **114** in a slightly upward direction. The transfer bed **114** is pivotable about a horizontal axis at an upstream edge or end that is adjacent the upstream tray conveyor **104** and the article conveyor **110**. The article conveyor **110** is oriented in a substantially horizontal plane and the downstream tray conveyor **108** is oriented at an oblique angle relative to horizontal, and optionally at an identical angle as the upstream tray conveyor **104**. The transfer bed **114** is pivotable between a horizontal or level orientation as best shown in FIG. **3** and an oblique or angled orientation as best shown in FIG. **2**. In the horizontal orientation, the transport surface **114***a* of the transfer bed **114** is oblique the conveyor surfaces **104***a* and **108***a* of the upstream and downstream tray conveyors **104** and **108** and is generally parallel and subjacent to the conveyor surface **110***a* of the article conveyor **110** (FIG. **3**). In the oblique orientation, the transport surface **114***a* is transfer bed **114** is generally parallel and coplanar with the conveyor surfaces **104***a* and **108***a* of the upstream and downstream tray conveyors **104** and **108** (FIG. **2**).
(20) The transfer bed **114** includes a lift or lifting apparatus, in the form of a plurality of stationary or fixed monoliths, blocks, or support elements that extend or reach through openings or voids in the transport surface **114***a* of the transfer bed **114** when the transfer bed is in the horizontal orientation. The lift apparatus is provided to lift or otherwise position the transport plate **124** of a tray **106** supported on the transfer bed **114** at a receiving position that is substantially coplanar to the conveyor surface **110***a* of the article conveyor **110**. The transfer bed **114** pivots relative to the support elements such that as the transfer bed **114** is lowered to the horizontal orientation, the support elements extend at least partially above the transport surface **114***a* of the transfer bed **114**. In this manner, as the transfer bed **114** and the tray **106** supported thereon lower, the support elements contact the plate **124** of the tray **106** and extend or reach through the aperture **122** of the frame **116** and the frame **116** continues to lower relative to the transport plate **124**. The transfer bed **114** may include a tray stop or tray positioning apparatus for positioning the tray in an acceptable position such that the support elements may engage the transport plate **124** of the tray **106** without engaging, colliding, or interfering with the frame **116** while the support elements reach/extend through the aperture **122**. The lift apparatus of transfer bed **114** may be similar or substantially identical in structure and function to the exemplary de-traying apparatus disclosed in previously mentioned U.S. Patent Application Pub. No. 2022/0055843A1, the disclosure of which is hereby incorporated herein by reference in its entirety.
(21) The support elements are positioned such that when the transport plate **124** is wholly supported by and resting on the support elements the transport plate **124** is positioned in the receiving position in which the transport surface **124***a* of the plate is horizontal and substantially parallel with the conveyor surface **110***a* of the article conveyor **110**. The height of the transport surface **124***a* of the transport plate **124** in the receiving position is set in order to reduce or substantially eliminate impact forces to articles **112** as they transfer from the transport surface **110***a* of article conveyor **110** to the tray **106**. Optionally, the height difference between the transport surface **124***a* of the transport plate **124** and the conveyor surface **110***a* of the article conveyor **110** is about ten millimeters (10 mm) or less such that the article **112** drops ten millimeters or less to the tray **106**. Optionally, the height difference is about five millimeters (5 mm) or less such that the article **112** drops five millimeters or less to the tray **106**. In the receiving position, the transport surface **124***a* of the transport plate **124** is at or above the raised edge **118** of the frame **116** of the tray **106**. With the transfer bed **114** in the oblique orientation, a tray **106** supported at the transfer bed **114** is positioned in a transport position in which the transport plate **124** of the tray is supported on the base **120** of the frame **116** of the tray **106** (i.e. the transport surface **124***a* of the plate is below the raised edge **118** of the frame **116**).

(22) The tray loading system **100** is operable to load a tray **106** in the following manner. A tray **106** is conveyed along the upstream tray conveyor **104** and then transferred from the conveyor surface **104***a* to the transfer bed **114** with the transfer surface **114***a* in the oblique orientation. Once the tray is substantially supported on the transfer bed **114**, the transfer bed **114** pivots downward from the oblique position shown in FIG. **2** to the horizontal position shown in FIG. **3**. While lowering, the transport plate **124** engages or comes to rest on the fixed support elements of the lifting apparatus and the tray frame **116** continues to lower. With the transport plate **124** in the receiving position, the article conveyor **110** transports and then transfers an article **112** from the conveyor surface **110***a* to the transport surface **124***a* of the **106**. With the article supported at the transport plate **124**, the transfer bed **114** raises from the horizontal orientation to the oblique orientation and the base **120** of the tray frame **116** engages the transport plate **124**. With the tray **106** in the transport position the transfer bed **114** transfers the tray **106** with the article **112** from the transport surface **114***a* to the transport surface **108***a* of the downstream tray conveyor **108** which then transports the tray away from the transfer bed **114**. The transfer bed **114** is then ready to receive an empty tray **106** from the upstream conveyor **104** for loading of another article **112**.
(23) Referring to the illustrative embodiments of FIGS. **8-9**, a tray loading system **200** is provided for a material handling system **202** having a first or upstream tray conveyor **204** for transporting material handling transport trays **106** (as previously described above and illustrated in FIGS. **5-7**), a second or downstream tray conveyor **208** for transporting trays **106**, and an upstream article conveyor **210** for transporting articles or items **112**. The tray loading system **200** includes a transfer bed **214** for facilitating a transfer of an article **112** from the article conveyor **210** to a tray **106** supported at the transport surface **214***a* of the transfer bed **214**. The upstream tray conveyor **204** delivers trays **206** to the transfer bed **214** and the downstream tray conveyor **208** transports trays **106** away from the transfer bed **214**.
(24) The transfer bed **214** is positioned downstream of the upstream tray conveyer **204** and includes a transport or support surface **214***a* for supporting the frame **216** of a tray **106** at the transfer bed **214**. In the illustrated embodiment of FIGS. **8-9**, the upstream tray conveyor **204**, downstream tray conveyor **208**, and article conveyor **210** are each oriented horizontally. The transport surface **214***a* of transfer bed **214** is generally parallel and coplanar with the conveyor surfaces **204***a* and **208***a* of the upstream and downstream tray conveyors **204** and **208** (FIG. **8**).
(25) The transfer bed **214** includes a lift apparatus, in the form of a plurality of vertically movable support elements **226** that are raiseable to extend or reach through openings or voids in the transport surface **214***a* of the transfer bed **214** to engage the transport plate **124** of the tray **106** through the aperture **122** (FIG. **8**). The support elements **226** are provided to lift the transport plate **124** of a tray **106** supported on transfer bed **214** to a receiving position that is substantially coplanar to the conveyor surface **210***a* of the article conveyor **210** (FIG. **8**). As the support elements **226** raise they lift the transport plate **124** of the tray **106** above the peripheral raised edge **118** of the tray **106** and the frame **116** of the tray remains on the transport surface **214***a* of the transfer bed **214**.
(26) The support elements are operable to position the transport plate **124** at the receiving position in which the transport surface **124***a* of the plate is horizontal and at or about coplanar with the conveyor surface **210***a* of the article conveyor **210**. The height of the transport surface **124***a* of the transport plate **124** in the receiving position is set in order to reduce or substantially eliminate impact forces to articles **112** as they transfer from the transport surface **210***a* of article conveyor **210** to the tray **106**. Optionally, the height difference between the transport surface **124***a* of the transport plate **124** and the conveyor surface **210***a* of the article conveyor **210** is about ten millimeters (10 mm) or less such that the article **112** drops ten millimeters or less to the tray **106**. Optionally, the height difference is about five millimeters (5 mm) or less such that the article **112** drops five millimeters or less to the tray **106**. In the receiving position, the transport surface **124***a* of the transport plate **124** is substantially above the raised edge **118** of the frame **116** of the tray **106**. With the lifting apparatus **226** at transfer bed **214** lowered, a tray **106** supported at the transfer

bed **214** is positioned in a transport position in which the transport plate **124** of the tray is supported on the base **120** of the frame **116** (i.e. the transport surface **124***a* of the plate is below the raised edge **118** of the frame **116**). The lift apparatus of transfer bed **214** may be similar or substantially identical in structure and function to the exemplary de-traying apparatus disclosed in previously mentioned U.S. Patent Application Pub. No. 2022/0055843A1, the disclosure of which is hereby incorporated herein by reference in its entirety.

(27) The tray loading system **200** is operable to load a tray **106** in the following manner. A tray **106** is conveyed along the upstream tray conveyor **204** and then transferred from the conveyor surface **204***a* to the transfer bed **214** with the transfer surface **214***a* substantially coplanar with the upstream tray conveyor. Once the tray is substantially supported on the transfer bed **214**, the support elements **226** of the lifting apparatus raise upward and engage the transport plate **124** through the aperture **122** while the tray frame **116** remains at the transport surface **214***a* of the transfer bed **214**. With the transport plate **124** in the receiving position, the article conveyor **210** transports and then transfers an article **112** from the conveyor surface **210***a* to the transport surface **124***a* of the tray **106**. With the article supported at the transport plate **124**, the support elements **226** lower down and the transport plate **124** lowers until it engages or rests on the base **120** of the tray frame **116**. With the tray **106** in the transport position the transfer bed **214** transfers the tray **106** with the article **112** from the transport surface **214***a* to the conveyor surface **208***a* of the downstream tray conveyor **208** which then transports the tray away from the transfer bed **214**. The transfer bed **214** is then ready to receive an empty tray **106** from the upstream conveyor **204** for loading of another article **112**.

(28) Referring to the illustrative embodiments of FIGS. **10-11**, a tray loading system **300** is provided for a material handling system **302** having a first or upstream tray conveyor **304** for transporting material handling transport trays **106** (as previously described above and illustrated in FIGS. **5-7**), a second or downstream tray conveyor **308** for transporting trays **106**, and an upstream article conveyor **310** for transporting articles or items **112**. The tray loading system **300** includes similar components as the tray loading system **200** described above and illustrated in FIGS. **8** and **9**, and functions in a substantially similar manner as tray loading system **200**, with significant differences discussed below. Tray loading system **300** includes a transfer bed **314** for facilitating a transfer of an article **112** from the article conveyor **310** to a tray **106** supported at the transport surface **314***a* of the transfer bed **314**. The upstream tray conveyor **304** delivers trays **306** to the transfer bed **314** and the downstream tray conveyor **308** transports trays **106** away from the transfer bed **314**.

(29) The transfer bed **314** is positioned downstream of the upstream tray conveyer **304** and includes a transport or support surface **314***a* for supporting the frame **316** of a tray **106** at the transfer bed **314**. In the illustrated embodiment of FIGS. **10-11**, the upstream tray conveyor **304** is oriented at an oblique angle relative to horizontal, as shown in FIG. **10**, and the downstream tray conveyor **308** and article conveyor **310** are each oriented horizontally. The transport surface **314***a* of transfer bed **314** is generally at about an equal height with the discharge end or the conveyor surface **304***a* of the upstream tray conveyor **304** and is generally parallel and coplanar with the conveyor surface **308***a* of the downstream tray conveyor **308** (FIG. **10**). The discharge end of the article conveyor **310** is positioned substantially superjacent to the discharge end of the upstream tray conveyor **304**. In comparison to the relatively large height difference between conveyor surfaces **204***a* and **210***a* of tray loading system **200**, the height difference between the discharge ends of conveyor surfaces **304***a* and **310***a* of tray loading system **300** is relatively small, as illustrated in FIG. **10**. As such, enough space is provided for a tray **106** to pass freely between the conveyor surface **304***a* and a support frame of the article conveyor **310** to the transfer bed **314** without requiring a significant height difference between the upstream tray conveyor **304** and article conveyor **310**.

(30) The transfer bed **314** includes a lift apparatus, in the form of a plurality of vertically movable support elements **326** that are raiseable to extend or reach through openings or voids in the transport surface **314***a* of the transfer bed **314** to engage the transport plate **124** of the tray **106**

through the aperture **122** (FIG. **10**). The support elements **326** are provided to lift the transport plate **124** of a tray **106** supported on transfer bed **314** to a receiving position that is substantially coplanar to the conveyor surface **310***a* of the article conveyor **310** (FIG. **10**). As the support elements **326** raise, they lift the transport plate **124** of the tray **106** above the peripheral raised edge **118** of the tray **106** and the frame **116** of the tray remains on the transport surface **314***a* of the transfer bed **314**. In comparison to the movement length of the vertically movable support elements **226** of tray loading system **200**, the vertically movable support elements **326** of tray loading system **300** are movable through a significantly shorter length, as the movement requirements for raising the transport plate **124** of the tray to the height of the conveyor surface **310***a* of article conveyor **310** is significantly shorter.

(31) The support elements are operable to position the transport plate **124** at the receiving position in which the transport surface **124***a* of the plate is horizontal and at or about coplanar with the conveyor surface **310***a* of the article conveyor **310**. The height of the transport surface **124***a* of the transport plate **124** in the receiving position is set in order to reduce or substantially eliminate impact forces to articles **112** as they transfer from the transport surface **310***a* of article conveyor **310** to the tray **106**. Optionally, the height difference between the transport surface **124***a* of the transport plate **124** and the conveyor surface **310***a* of the article conveyor **310** is about ten millimeters (10 mm) or less such that the article **112** drops ten millimeters or less to the tray **106**. Optionally, the height difference is about five millimeters (5 mm) or less such that the article **112** drops five millimeters or less to the tray **106**. In the receiving position, the transport surface **124***a* of the transport plate **124** is substantially above the raised edge **118** of the frame **116** of the tray **106**. With the lifting apparatus **326** at transfer bed **314** lowered, a tray **106** supported at the transfer bed **314** is positioned in a transport position in which the transport plate **124** of the tray is supported on the base **120** of the frame **116** (i.e. the transport surface **124***a* of the plate is below the raised edge **118** of the frame **116**). The lift apparatus of transfer bed **314** may be similar or substantially identical in structure and function to the exemplary de-traying apparatus disclosed in previously mentioned U.S. Patent Application Pub. No. 2022/0055843A1, the disclosure of which is hereby incorporated herein by reference in its entirety.

(32) The tray loading system **300** is operable to load a tray **106** in the following manner. A tray **106** is conveyed along the upstream tray conveyor **304** and then transferred from the conveyor surface **304***a* to the transfer bed **314** with the tray **106** traveling in slightly upward direction along the upstream tray conveyor **304** and then transferring to the horizontal transport surface **314***a* of the transfer bed **314**. Once the tray is substantially supported on the transfer bed **314**, the support elements **326** of the lifting apparatus raise upward and engage the transport plate **124** through the aperture **122** while the tray frame **116** remains at the transport surface **314***a* of the transfer bed **314**. With the transport plate **124** lifted into the receiving position, the article conveyor **310** transports and then transfers an article **112** from the conveyor surface **310***a* to the transport surface **124***a* of the tray **106**. With the article supported at the transport plate **124**, the support elements **326** lower down and the transport plate **124** lowers until it engages or rests on the base **120** of the tray frame **116**. With the tray **106** in the transport position, the transfer bed **314** transfers the tray **106** with the article **112** from the transport surface **314***a* to the conveyor surface **308***a* of the downstream tray conveyor **308** which then transports the tray away from the transfer bed **314**. The transfer bed **314** is then ready to receive an empty tray **106** from the upstream conveyor **304** for loading of another article **112**.

(33) Referring to the illustrative embodiments of FIGS. **12-13**, and **13**A, a tray loading system **400** is provided for a material handling system **402** having a first or upstream tray conveyor **404** for transporting material handling transport trays **106** (as previously described above and illustrated in FIGS. **5-7**), a second or downstream tray conveyor **408** for transporting trays **106**, and an upstream article conveyor **410** for transporting articles or items **112**. The tray loading system **400** includes similar components as the tray loading systems **200** and **300** described above and illustrated in

FIGS. **8-9** and **10-11**, and functions in a substantially similar manner as tray loading systems **200** and **300**, with the differences discussed below. The tray loading system **400** includes a transfer bed **414** for facilitating a transfer of an article **112** from the article conveyor **410** to a tray **106** supported at the transport surface **414***a* of the transfer bed **414**. The upstream tray conveyor **404** delivers trays **406** to the transfer bed **414** and the downstream tray conveyor **408** transports trays **106** away from the transfer bed **414**.

(34) The transfer bed **414** is positioned downstream of the upstream tray conveyer **404** and includes a transport or support surface **414***a* for supporting the frame **416** of a tray **106** at the transfer bed **414**. In the illustrated embodiment of FIGS. **12-13**, the upstream tray conveyor **404**, downstream tray conveyor **408**, and article conveyor **410** are each oriented horizontally. The transport surface **414***a* of transfer bed **414** is generally parallel and coplanar with the conveyor surface of the upstream tray conveyor **404** and downstream tray conveyor **408** to smoothly (i.e. with little or no impact and/or vertical shifting) receive a tray **106** from the upstream tray conveyor **404** and discharge or transfer a tray **106** to the downstream tray conveyor **408**. The upstream tray conveyor **404**, transfer bed **414**, and downstream tray conveyor **408** are each co-linear with each other, i.e. their conveyance directions are the same. The article conveyor **410** is positioned alongside and parallel to transfer bed **414** and parallel to the upstream tray conveyor **404** and downstream tray conveyor **408**.

(35) The transfer bed **414** includes a lift apparatus, in the form of a plurality of vertically movable support elements **426** that are raiseable to extend or reach through openings or voids in the transport surface **414***a* of the transfer bed **414** to engage the transport plate **124** of the tray **106** through the aperture **122**, as described above. Vertically movable support elements **426** function similar to that described previously for support elements **226** and **326**. The support elements **426** are provided to lift the transport plate **124** of a tray **106** supported on transfer bed **414** to a receiving position that is substantially coplanar to the conveyor surface **410***a* of the adjacent article conveyor **410**. As the support elements **426** raise they lift the transport plate **124** of the tray **106** above the peripheral raised edge **118** of the tray **106** and the frame **116** of the tray remains on the transport surface **414***a* of the transfer bed **414**. As best seen in FIG. **13**, the tray loading system **400** includes a lateral transfer apparatus including a pair of pushers **428** operable to push articles **112** laterally from the conveyor surface **410***a* of the article conveyor **410** onto the transport surface **124***a* of the transport plate **124** of a tray **106** supported at the transfer bed **414**.

(36) Similar to the function of transfer bed **314** and the support elements **326** of tray loading system **300**, the support elements **426** of system **400** are operable to position the transport plate **124** at the receiving position in which the transport surface **124***a* of the plate is at or about coplanar with the conveyor surface **410***a* of the article conveyor **410**. The height of the transport surface **124***a* of the transport plate **124** in the receiving position is set in order to reduce or substantially eliminate impact forces to articles **112** as they transfer from the transport surface **410***a* of article conveyor **410** to the tray **106**. Optionally, the height difference between the transport surface **124***a* of the transport plate **124** and the conveyor surface **410***a* of the article conveyor **410** is about ten millimeters (10 mm) or less such that the article **112** drops ten millimeters or less to the tray **106**. Optionally, the height difference is about five millimeters (5 mm) or less such that the article **112** drops five millimeters or less to the tray **106**. In the receiving position, the transport surface **124***a* of the transport plate **124** is substantially above the raised edge **118** of the frame **116** of the tray **106**. With the lifting apparatus **426** at transfer bed **414** lowered, a tray **106** supported at the transfer bed **414** is positioned in a transport position in which the transport plate **124** of the tray is supported on the base **120** of the frame **116** (i.e. the transport surface **124***a* of the plate is below the raised edge **118** of the frame **116**). The lift apparatus of transfer bed **414** may be similar or substantially identical in structure and function to the exemplary de-traying apparatus disclosed in previously mentioned U.S. Patent Application Pub. No. 2022/0055843A1, the disclosure of which is hereby incorporated herein by reference in its entirety.

(37) The tray loading system **400** is operable to load a tray **106** in the following manner. A tray **106** is conveyed along the upstream tray conveyor **404** and then transferred from the conveyor surface **404***a* to the transfer bed **414** with the transfer surface **414***a* substantially coplanar with the conveyor surface of the upstream tray conveyor **404**. Once the tray **106** is substantially supported on the transfer bed **414**, the support elements **426** of the lifting apparatus raise upward and engage the transport plate **124** through the aperture **122** while the tray frame **116** remains at the transport surface **414***a* of the transfer bed **414**. The article conveyor **410** transports an article **112** to a position adjacent or proximate the transfer bed **414**. With the transport plate **124** in the receiving position, one of the pair of pushers **428** pushes the article **112** from the conveyor surface **410***a* of article conveyor **410** to the transport surface **124***a* of the tray **106**. With the article supported at the transport plate **124**, the support elements **426** lower down and the transport plate **124** lowers until it engages or rests on the base **120** of the tray frame **116**. With the tray **106** in the transport position, the transfer bed **414** transfers the tray **106** with the article **112** from the transport surface **414***a* to the conveyor surface **408***a* of the downstream tray conveyor **408** which then transports the tray away from the transfer bed **414**. The transfer bed **414** is then ready to receive an empty tray **106** from the upstream conveyor **404** for loading of another article **112**.
(38) Referring now to the illustrative embodiment of FIG. **14**, a method **500** is provided for loading an article **112** (e.g. a fragile product or item such as fresh tomatoes or glass light bulbs) onto a material handling transport tray **106** in a material handling conveyor system, such as material handling conveyor systems **102**, **202**, **302**, or **402** described above and illustrated in FIGS. **1-4**, **8-9**, **10-11**, and **12-13**A. Unless specifically noted otherwise, method **500** is hereinafter described in relation to tray loading system **100** of the illustrative embodiment of FIGS. **1-4**. The method **500** includes providing **502** a first or upstream tray conveyor **104**, an article conveyor **110** adjacent the first tray conveyor, a tray **106** (see FIGS. **5-7**) including a frame **116** having a peripheral raised edge **118** and a base **120** having at least one aperture **122**, and a transport plate **124** with a transport surface **124***a* for supporting articles **112**. The transport plate **124** is supportable on the base **120** of the frame **116** for transport and is moveable relative to the raised edge **118** and base **120** of the frame **116**. A transfer bed **114** is provided at **502** to receive the tray **106** from the upstream tray conveyor **104** and is operable to position the transport plate **124** of the tray **106** at a receiving position to receive an article **112** from the conveyor surface **110***a* of the article conveyor **110**, optionally with a drop of less than about ten millimeters (10 mm) and optionally less than about five millimeters (5 mm). Method **500** includes delivering **504** a tray with the upstream tray conveyor to the transfer bed **114** and positioning **506** the transport plate **124** of the tray **106** at the transfer bed **114** such that the transport surface **124***a* of the transport plate **124** is less than about ten millimeters (10 mm) (or less than about five millimeters (5 mm)) below the conveyor surface **110***a* of the article conveyor **110**. An article **112** is transferred **508** from the conveyor surface **110***a* of the article conveyor **110** to the transport surface **124***a* of the transport plate **124** of the tray **106** at the transfer bed **114**. Once the article **112** is loaded onto the tray, the loaded tray is transferred **510** away from the transfer bed **114**. Optionally, a second or downstream tray conveyor **108** is provided downstream of the transfer bed **114** to transfer **510** the loaded tray away from the transfer bed. Positioning the transport plate of the tray at **506** may include operating a lift apparatus to move either the frame **116** of the tray **106** or the transport plate **124** of the tray **106** relative to the other.
(39) When method **500** is performed with tray loading systems similar to tray loading system **100** of the illustrative embodiment of FIGS. **1-4**, positioning the transport plate **124** includes pivoting or lowering the transfer bed **114** downward from its oblique orientation to its horizontal orientation. Accordingly, lowering the frame **116** of the tray **106** relative to the support elements permits the frame **116** to lower relative to the transport plate **124** of the tray. The transport plate **124** remains in substantially the same position relative to the support elements.
(40) When method **500** is performed with tray loading systems similar to tray loading system **200**, **300**, or **400** of the illustrative embodiments of FIG. **8-9**, **10-11**, or **12-13**, positioning the transport

plate **124** includes operating the lift apparatus to move the transport plate **124** to a receiving position such that the transport surface **124***a* is substantially coplanar to the conveyor surface **110***a* of the article conveyor **110** (or less than about five millimeters (5 mm)). Optionally, the transport plate **124** is positioned to receive an article **112** from the article conveyor **110** with less than about ten millimeters (10 mm) (or less than about five millimeters (5 mm)) drop to the transport plate **124**. Operating the lift apparatus may include raising the plurality of support elements **226**, **326**, or **426** such that they lift the transport plate **124** of the tray **106** such that the transport surface **124***a* is at or above the peripheral raised edge **118** of the tray and the frame **116** of the tray remains substantially on the transport surface **114***a* of the transfer bed **114**.
(41) When method **500** is performed with tray loading systems similar to tray loading system **400** of the illustrative embodiment of FIGS. **12-13**, the transferring **508** an article from the conveyor surface **410***a* of the article conveyor **410** to the transport surface **124***a* of the transport plate **124** includes pushing the article **112** with one of the pushers **428** from the conveyor surface **410***a* onto the transport surface **124***a*.
(42) Thus, the tray loading systems **100**, **200**, **300**, and **400**, and the method **500** are especially suited for loading fragile articles onto transport trays within a material handling conveyor system, though they are not so limited. The tray loading systems include transfer beds with lifting systems, which position transport plates of the trays near the height of the conveyor surface of the article conveyor in a manner that reduces or substantially eliminates dropping forces when the articles transfer from the article conveyor onto the tray.
(43) Changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the present invention, which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

Claims

1. A tray loading system for loading articles onto trays in a material handling conveyor system, said tray loading system comprising: a first tray conveyor having a conveyor surface and being operable to transport a material handling transport tray in a conveyor direction for loading, wherein said material handling transport tray comprises: a frame having a peripheral raised edge and a base having at least one aperture; and a transport plate with a transport surface for supporting articles, wherein said transport plate is supportable on said base of said frame for transport and is moveable relative to said raised edge and said base of said frame; a transfer bed adjacent and downstream of said first tray conveyor and having a transport surface configured to receive said material handling transport tray from said first tray conveyor; an article conveyor adjacent said transfer bed and said first tray conveyor, said article conveyor having a conveyor surface and being operable to transport an article in the conveyor direction and to transfer the article from said conveyor surface of said article conveyor onto said transport surface of said material handling transport tray when supported at a portion of said transfer bed; a second tray conveyor downstream of said transfer bed having a conveying surface and being operable to transport said material handling transport tray in the conveyor direction away from said transfer bed; said transfer bed operable to position said material handling transport tray in a receiving position wherein said transport plate of said material handling transport tray is at or above said raised edge of said frame of said material handling transport tray and such that said transport surface of said transport plate is substantially parallel with said conveyor surface of said article conveyor; and said transfer bed operable to position said material handling transport tray in a transport position wherein said transport plate of said material handling transport tray is supported on said base of said frame.
2. The tray loading system of claim 1, wherein said transport surface of said transport bed is substantially parallel and coplanar with said conveyor surfaces of said first and second tray

conveyors and is substantially parallel and subjacent to said conveyor surface of said article conveyor.

3. The tray loading system of claim 2, wherein said transfer bed comprises a lift apparatus operable to lift said transport plate of said material handling transport tray supported on said transfer bed to a position substantially coplanar to the conveyor surface of said article conveyor such that said transport plate is positioned to receive the article from said article conveyor with ten millimeters (10 mm) or less drop to said transport plate.

4. The tray loading system of claim 3, wherein said lift apparatus comprises a plurality of vertically moveable support elements configured to reach through the transport surface of said transfer bed and the aperture in said material handling transport tray such that as said plurality of support elements raise they lift said transport plate of said material handling transport tray supported at said transfer bed above said peripheral raised edge of said tray and said frame of said material handling transport tray remains on said transport surface of said transfer bed.

5. The tray loading system of claim 1, further comprising a pusher operable to push articles laterally from said conveyor surface of said article conveyor onto said transport surface of said transport plate of said material handling transport tray supported at said transfer bed.

6. A tray loading system for loading articles onto trays in a material handling conveyor system, said tray loading system comprising: a first tray conveyor operable to transport a material handling transport tray in a conveyor direction for loading, said material handling transport tray comprising: a frame having a peripheral raised edge and a base having at least one aperture; and a transport plate with a transport surface for supporting articles, wherein the transport plate is supportable on said base of said frame for transport and is moveable relative to said raised edge and said base of said frame; a transfer bed downstream of said first tray conveyor and configured to receive said material handling transport tray from said first tray conveyor; an article conveyor upstream of said transfer bed and superjacent to said first tray conveyor, said article conveyor operable to transport an article in the conveyor direction and to transfer the article from the conveyor surface of said article conveyor onto the transport surface of the material handling transport tray when supported at a portion of said transfer bed; and a second tray conveyor downstream of said transfer bed and operable to transport said material handling transport tray in the conveyor direction away from said transfer bed; said transfer bed operable to position the tray in a substantially level orientation wherein the transport plate of the material handling transport tray is at or above the raised edge of the frame of the material handling transport tray and such that the transport surface of the transport plate is substantially parallel with the conveyor surface of said article conveyor; and said transfer bed operable to position the material handling transport tray such that the transport plate of the material handling transport tray is substantially parallel to a conveyor surface of the downstream said second tray conveyor that is operable to transport the material handling transport tray from said transfer bed.

7. The tray loading system of claim 6, wherein said transfer bed is pivotable about a horizontal axis and operable to move between (i) a horizontal orientation in which the transport plate of material handling transport tray supported at said transfer bed is horizontal and substantially parallel with the conveyor surface of said article conveyor to receive the article from said article conveyor with 10 millimeters (10 mm) or less drop to the transport plate, and (ii) an oblique orientation in which the transport plate of said material handling transport tray supported at said transfer bed is obliquely oriented relative to horizontal and substantially parallel with the conveyor surface of said second tray conveyor to transfer said material handling transport tray from said transfer bed to said second tray conveyor.

8. The tray loading system of claim 6, wherein said transfer bed comprises a lift apparatus configured to position the transport plate of a horizontally orientated said material handling transport tray supported on a transport surface of said transfer bed at or above the peripheral raised edge of the frame of the material handling transport tray.

9. The tray loading system of claim 8, wherein said lift apparatus comprises a plurality of support elements fixed relative to said first tray conveyor and said article conveyor and configured to reach through the transport surface of said transfer bed and the aperture in the material handling transport tray as the frame of the material handling transport tray is lowered relative to said support elements such that the frame lowers relative to said transport plate of the material handling transport tray.
10. The tray loading system of claim 6, wherein a transport surface of said transfer bed is positioned substantially parallel with the conveyor surface of said first tray conveyor to receive said material handling transport tray from said first tray conveyor.
11. The tray loading system of claim 10, wherein said transfer bed comprises a lift apparatus operable to lift the transport plate of a horizontally orientated said material handling transport tray supported on said transfer bed to a position substantially coplanar to the conveyor surface of said article conveyor such that said transport plate is positioned to receive the article from said article conveyor with 10 millimeters (10 mm) or less drop to the transport plate.
12. The tray loading system of claim 11, wherein said lift apparatus comprises a plurality of vertically moveable support elements configured to reach through the transport surface of said transfer bed and the aperture in the material handling transport tray such that as said plurality of support elements raise they lift the transport plate of the material handling transport tray supported at said transfer bed above the peripheral raised edge of the tray and the frame of the tray remains on the transport surface of said transfer bed.
13. The tray loading system of claim 6, wherein the conveyor surfaces of said first tray conveyor and said article conveyor are each substantially horizontal.
14. The tray loading system of claim 6, wherein the conveyor surface of at least one of said first tray conveyor and said article conveyor is obliquely oriented relative to horizontal.
